

CAMBRIDGE
INTERNATIONAL EXAMINATIONS

NOVEMBER 2001

ADVANCED SUBSIDIARY LEVEL

MARK SCHEME

MAXIMUM MARK : 50

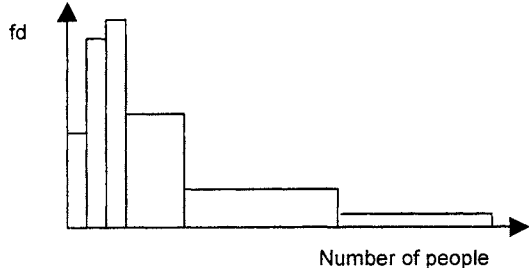
SYLLABUS/COMPONENT : 8709/6

MATHEMATICS



UNIVERSITY of CAMBRIDGE
Local Examinations Syndicate

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1	$\Sigma x = 105$ mean = 13.1 sd = 2.76 $\Sigma x^2 = 1439$	B1 B1 B1 3	For $\Sigma x^2 = 1439$ For answer For answer
2 (a)	Number of ways is ${}_{10}P_6$ or $10 \times 9 \times 8 \times 7 \times 6 \times 5$ = 151200	B1 B1 2	May be implied
(b)	$4! \times 3!$ = 144	B1 B1 B1 3	For 4! For 3! For answer
3 (i)	$P(\text{receives message}) = 0.4 \times 0.6 + 0.5 + 0.1 \times 0.8$ = 0.82	M1 M1 A1 3	For two 2-factor terms For adding 0.5 For correct answer
(ii)	$P(\text{Email} \mid \text{Receives})$ = 0.293	B1 M1 A1 3	For correct expression for numerator For dividing by their 0.82 For correct answer
4 (i)	Class width 20, 20, 20, 40, 100, 100 Frequency density: 2.3, 5.5, 6.1, 2.5, 0.86, 0.36 	B1 M1 M1 A1 A1 5	For class widths Attempt at frequency density or scaled frequency Graph with 6 bars of appropriate relative widths (any height) For x-axis going from 0 – 300 properly All correct including axes labelled
(ii)	$\left(\frac{122 + 110 + 46}{500} \right)^3 = 0.172$	M1 A1 2	For cubing their probability For correct answer

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5	(i)	$z = \frac{10 - 15}{4.2} = -1.190$	M1	Standardising and using tables
			M1	For subtracting a probability from 1
		$P(X < 10) = \Phi(-1.190) = 1 - 0.883 = 0.117$	A1 3	For correct answer
	(ii)	$z = 1.282$	B1	For correct z-value
		$\frac{T - 15}{4.2} = 1.282$	M1	For an equation relating T and their z
		$T = 20.4$	A1 3	For correct answer
	(iii)	$P(z > 1.19) = 1 - \Phi(1.19) = 1 - 0.8830 = 0.117$	B1	For 0.883 seen (or symmetry)
		Number of people = $0.117 \times 200 (= 23.4)$	M1	For multiplying a probability by 200
		Answer = 23	A1 3	For correct answer 23
	6	(i)	M1	For calculating $P(10)$, $P(11)$, $P(12)$
			M1	For correct use of binomial coefficients
			A1	For correct numerical expression
			A1 4	For correct answer
		(ii)	B1	For both mean and variance correct
			M1	For correct standardising process with or without cc
			A1	For correct use of continuity correction
			M1	For correct use of tables
			A1 5	For correct answer

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7	(i)	EITHER $P(X=0) = \frac{7}{10} \times \frac{6}{9} \times \frac{5}{8} \times \frac{4}{7} = \frac{1}{6}$	M1	For multiplying 4 probabilities together										
		and $P(X=1) = \frac{3}{10} \times \frac{7}{9} \times \frac{6}{8} \times \frac{5}{7} \times 4 = \frac{1}{2}$	A1	For correct given answer										
			M1	For multiplying by 4										
			A1	For obtaining given answer legitimately										
			B2	For showing given answer legitimately										
	(ii)	OR ${}_7C_4 \div {}_{10}C_4 = 1/6$	B2	4										
		${}_7C_3 \times {}_3C_1 \div {}_{10}C_4 = 1/2$												
		<table border="1"><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>Prob</td><td>0.167</td><td>0.5</td><td>0.3</td><td>0.0333</td></tr></table>	X	0	1	2	3	Prob	0.167	0.5	0.3	0.0333	M1	For attempting to find $P(X=0,1,2,3)$
		X	0	1	2	3								
		Prob	0.167	0.5	0.3	0.0333								
	A1	For 0.3 or 3/10												
	A1	3	For 0.0333 or 1/30											
(iii)	$E(X) = 1.2$	M1	For $\sum x_j p_j$											
		A1	For correct answer (must be exact)											
	$\text{Var}(X) = \sum x_i^2 p_i - \text{their } 1.2^2$	M1	For $\sum x_i^2 p_i - \text{their } 1.2^2$											
	$= 0.56$	A1	4	For correct answer										